GENERAL

The purpose of the Dry Low NOx (DLNx) control system is to regulate the distribution of the fuel and the location of flame in the staged multi-nozzle combustors around the gas turbine. A diagram of a DLNx I combustor is provided in Figure A.

GAS FUEL SYSTEM

A schematic of the DLNx gas fuel delivery systems is shown in Figure B. The fuel system is made up of the gas fuel speed/ratio valve, the gas control valve, the gas splitter valve, and the gas transfer valve.

The speed/ratio valve (SRV) is designed to maintain a predetermined pressure, P2, at the control valve inlet. The gas control valve (GCV) regulates the desired overall gas fuel flow delivered to the turbine in response to the command signal FSR from the SPEEDTRONIC™ panel.

The total gas flow is separated into three stages by the DLNx I system valves, the gas splitter valve (GSV) and the gas transfer valve (GTV). The DLNx I valves control the percentage of the total fuel flow delivered to the different stages of the DLNx I multi-nozzle combustor.

The gas splitter valve (GSV) regulates the distribution of the fuel flow to the primary and secondary zones in the combustor. The gas transfer valve (GTV) divides the fuel supplied to the secondary zone between two separate flow paths in the secondary fuel nozzle. The secondary fuel nozzle assembly contains a flow passage that mixes air and fuel ("secondary" nozzle) and a normal diffusion type passage ("transfer" nozzle).

LIQUID FUEL SYSTEM (IF APPLICABLE)

The liquid fuel system, is composed of the liquid fuel stop valve, main fuel pump, control valve, fuel filter, flow divider, and the liquid fuel splitter valve. The fuel pump and control valve work in conjunction with the flow divider and a number of orifices in each secondary fuel line to supply each combustion chamber with the required fuel flow.

Liquid fuel can be supplied to a set of liquid fuel nozzles in the primary zone and a nozzle in the secondary zone of the DLNx I combustor. The liquid fuel splitter valve is an on/off type valve that is used to divert a fixed percentage of the total fuel flow down the secondary zone flow path. Typically, an orifice in the liquid fuel piping is sized to govern this fixed percentage through pressure flow division.

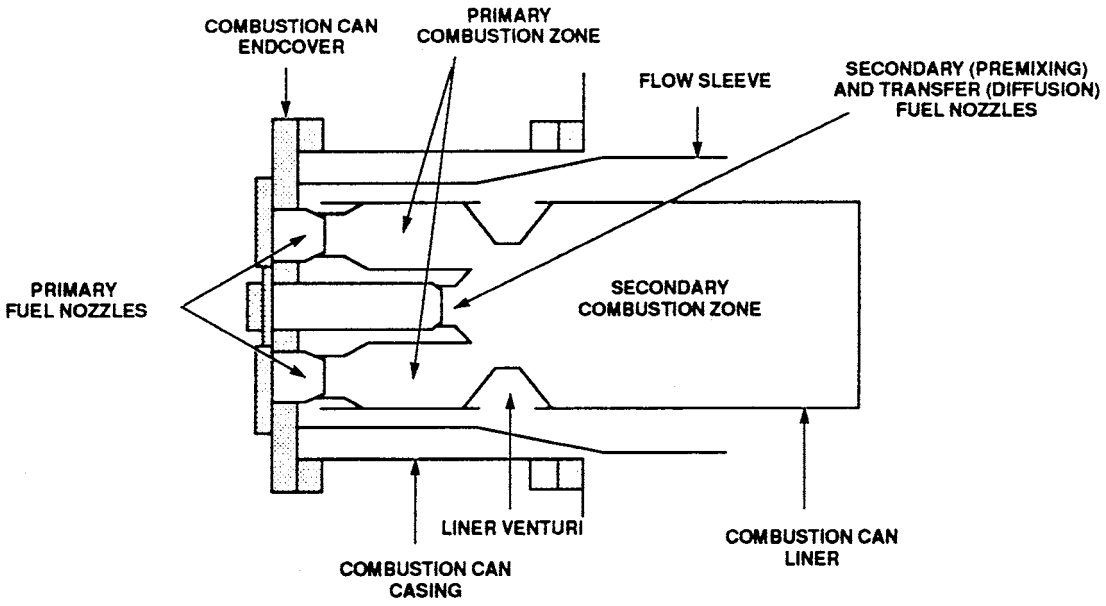


Figure A DLNx I Multi-Nozzle Staged Combustor Cross-Section

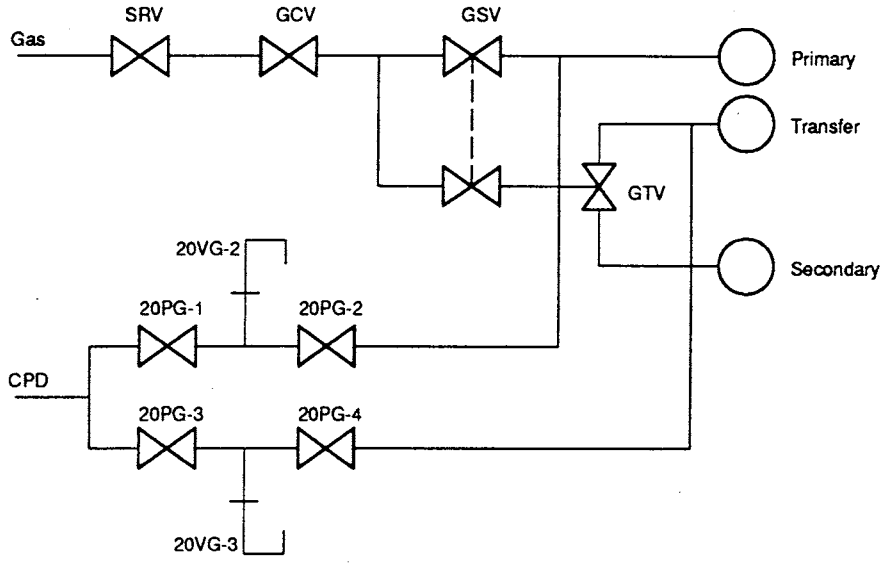


Figure B Gas Fuel Delivery System

FLAME DETECTION

Reliable detection of the flame location in the DLNx I system is critical to the control of the combustion process and to the protection of the gas turbine hardware. Four flame detectors in separate combustion chambers are mounted to detect primary zone flame. Four additional

flame detectors in separate combustion chambers are dedicated to detect secondary flame. The signals from these flame detectors are processed in control logic and used for various control and protection functions including the DLNx combustion mode determination.

IGNITION SYSTEM

Two spark plugs located in different combustion chambers are used to ignite the fuel flow in the primary zone of the combustors. These spark plugs are energized to ignite the fuel during startup and during certain DLNx I mode changes that require re-ignition of the primary combustion zone. Flame is propagated to those combustion chambers through crossfire tubes that connect the primary zones of adjacent combustion chambers around the gas turbine. Flame is introduced in the secondary zone of the combustion chambers by igniting the secondary fuel flow with flame from the primary zone.

GAS FUEL OPERATION

The gas turbine firing temperature is defined as the temperature at the inlet to the first stage turbine nozzle. Due to the extreme temperatures in this region of the gas turbine, it is impractical to measure the firing temperature directly. The SPEEDTRONIC™ control software calculates a turbine firing temperature through known thermodynamic relationships by measuring the gas turbine exhaust temperature, the compressor discharge pressure and the ambient temperature.

The fuel flow split between the DLNx I combustion chamber fuel nozzles and the location of the flame (DLN mode) is regulated as a function of the calculated turbine firing temperature signal (TTRF1).

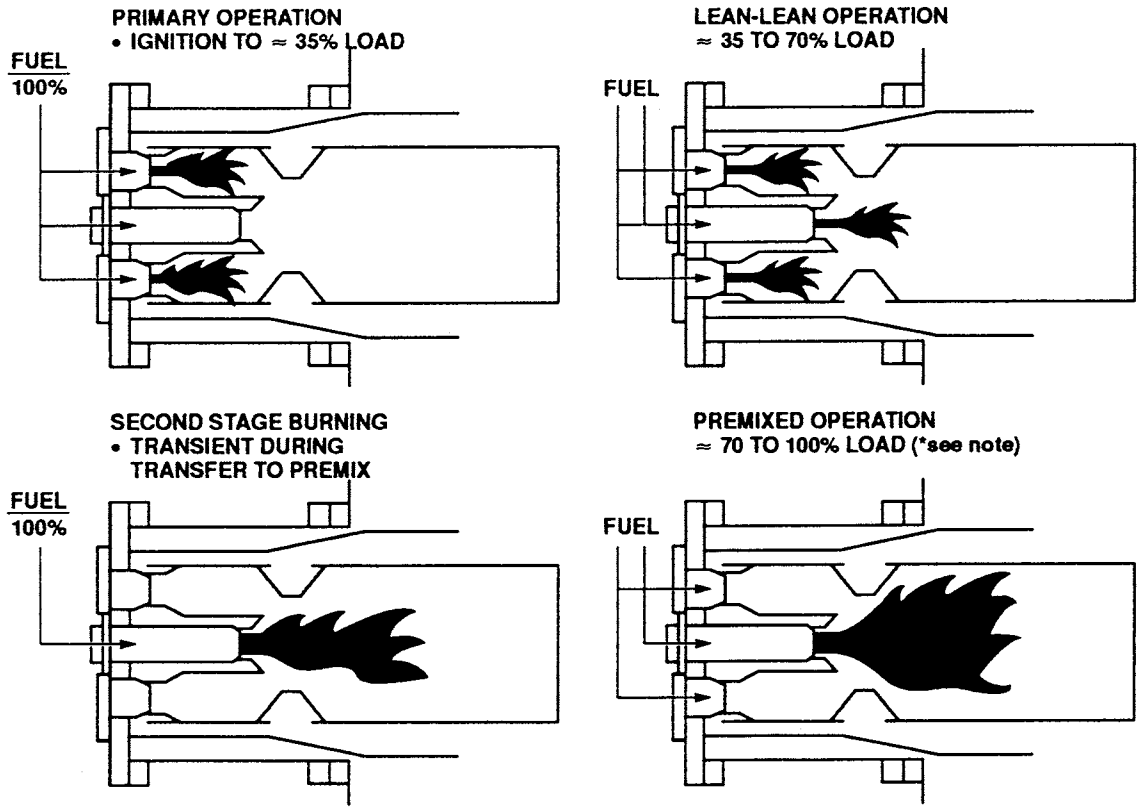
There are four basic modes of distributing the fuel to the DLNx I combustor (Figure B1).

Primary

Fuel and combustion are only in the primary zone. Primary combustion is from ignition to full speed then through approximately 35% load or 1500F* firing temperature.

Lean-lean

Fuel and combustion are in both the primary and secondary zones. Typical load is 35%-70% and firing temperature in lean-lean mode is from 1500F to 1850F*.



NOTE: Premixed Operation can be extended to lower load ranges with reduced Inlet Guide Vane Operation

Figure B1 Fuel Staged Dry Low NOx I Gas Fuel Operating Modes

Secondary

All fuel and combustion is transferred to the secondary zone of the combustor, through the transfer manifold. Secondary mode is a transitory mode to extinguish the primary flame during the transfer from Lean-lean to the Premix mode of operation. Typical firing temperature for this transition is 1850F*.

Premix

In this mode, fuel is in introduced in both the primary and secondary zones, with combustion only occurring in the secondary zone. Air and a majority of the fuel are mixed in the primary zone and combustion occurs only in the secondary zone. "Premixing" the air and fuel reduces gas turbine exhaust NOx emissions. This mode of operation is typically from 70%-100% load or above 1900F firing temperature.

Note: Refer to the Control Specifications for the correct firing temperature.

Figure C is a plot of the DLNx I operational modes as a function of load and firing temperature. During normal loading of a unit operating on gas fuel, the system will progress through Primary, Lean-Lean, Secondary and Premix modes. Normal unloading of the machine from high loads on gas fuel will go from Premix to Lean-lean after the spark plugs re-ignite the primary combustion zone. Further unloading will cause Primary mode operation to be initiated which will continue through breaker opening until turbine shutdown.

During the transfer from Lean-Lean to Premix, the gas transfer valve splits the gas fuel flow in the combustors' secondary stage fuel nozzle between the premixing "secondary" passage and the diffusion "transfer" passage. It is necessary to route some of the secondary fuel flow through the diffusion section of the secondary nozzle during this transfer period in order to lower combustion dynamics and stabilize the secondary flame.

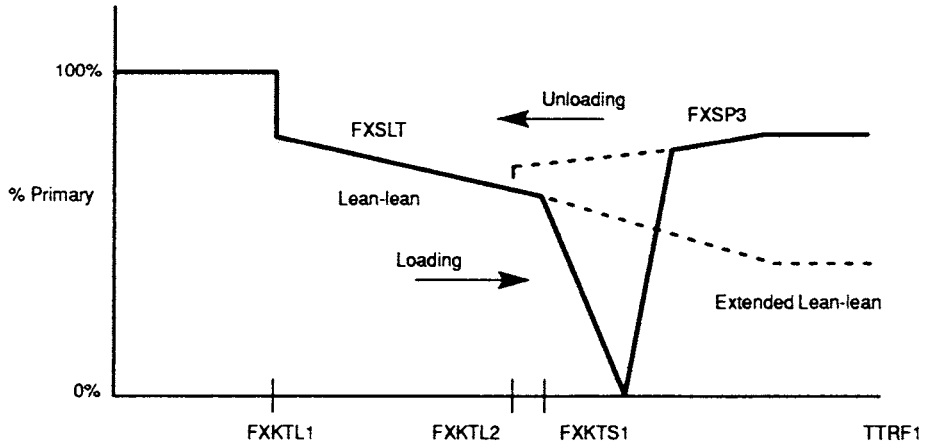


Figure C Liquid Fuel Splitter Valve Schedule

Figure C is a typical trace of the DLNx I valve positions during the transfer from Lean-lean to Premix mode*. The gas transfer valve will normally be in the 0% stroke position, meaning that 100% of the secondary fuel flow will be to the secondary premixing passage in the secondary nozzle assembly. During transfer to the Secondary mode of operation, flame is extinguished in the primary zone by the gas splitter valve diverting all flow to the secondary zone. In Secondary mode, to accommodate the higher flow of fuel flow through the secondary, the gas transfer valve will move to an intermediate position in order to route part of the secondary fuel flow through the diffusion passage.

After flame has been extinguished in the primary combustion zone, the machine transfers into the Premix mode of operation. The gas splitter valve moves to re-admit fuel into the primary zone and the gas transfer valve ramps to a position that will cause the secondary

fuel flow to be routed to the transfer diffusion passage of the secondary nozzle. After the gas splitter valve has reached its steady-state Premix mode position, the gas transfer valve will ramp to its normal position, diverting all of the secondary fuel flow through the premixing nozzle once again.

*Refer to the Control Specifications for the correct settings for a specific turbine.

LIQUID FUEL OPERATION (WHEN APPLICABLE)

In the DLNx I system, liquid fuel combustion occurs in either the Primary mode or the Lean-Lean mode. The DLNx I combustor is ignited during startup and operated in the Primary mode up to full speed no-load.

When the calculated firing temperature reaches a set value at some part load point, the liquid fuel shear valve will open and divert a fixed percentage of the total fuel flow down the liquid fuel passage in the secondary nozzle assembly. Diverting some of the total fuel flow into the secondary zone is necessary to prevent primary zone flame from damaging the combustion liner venturi which separates the primary and secondary zones.

Normal unloading of the turbine on liquid fuel from base load progresses at a constant split between primary and secondary fuel flow until a firing temperature setpoint causes the liquid fuel shear valve to shutoff the secondary fuel flow. The turbine is then unloaded and shutdown in the Primary mode.

*Refer to the Control Specifications for the correct settings for a specific turbine.

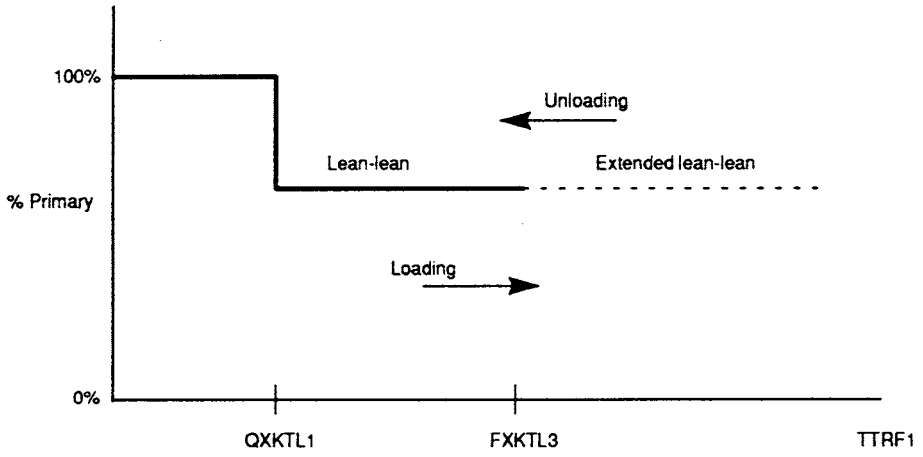


Figure D Liquid Fuel Splitter Valve Schedule

PURGE SYSTEM

With all DLN, the purge system is a critical part of the combustion and controls system. Purge air is used to cool the nozzles as well as keep back pressure from forcing fuel back into the manifolds where it could ignite and cause substantial damage to manifolds, piping or even the turbine. During fuel transfers and steady state operation, it is imperative that fuel not be allowed to enter an unused nozzle or manifold and ignite.

The purge system, which uses compressor discharge air on gas fuel and atomizing air for liquid fuel, consists of stop valves, limit and pressure switches, all of which provide feedback to the turbine control system. Because the purge system is connected directly to the fuel gas manifold, it is important to maintain a system of checks that will not allow the fuel to work backwards into the purge system and thus the compressor discharge casing, where it could ignite. These checks, on gas fuel purge, are accomplished through use "double block and bleed" purge system. Through the use of limit and pressure switches, the turbine control system ensures the correct position of the purge valves.

IGV TEMPERATURE CONTROL

With all DLN systems, calculated firing temperature is used to determine transfer points between operating modes. When calculating firing temperature (TFRFI) air flow through the turbine is one of the critical parameters in determining the maximum firing temperature. For this reason, the unit must be run with IGV temperature control turned ON, whenever operating in DLN mode. Running in DLN mode with IGV temperature control turned OFF, will lead to inaccurate firing temperature calculations and possibly overfiring of the combustion components.